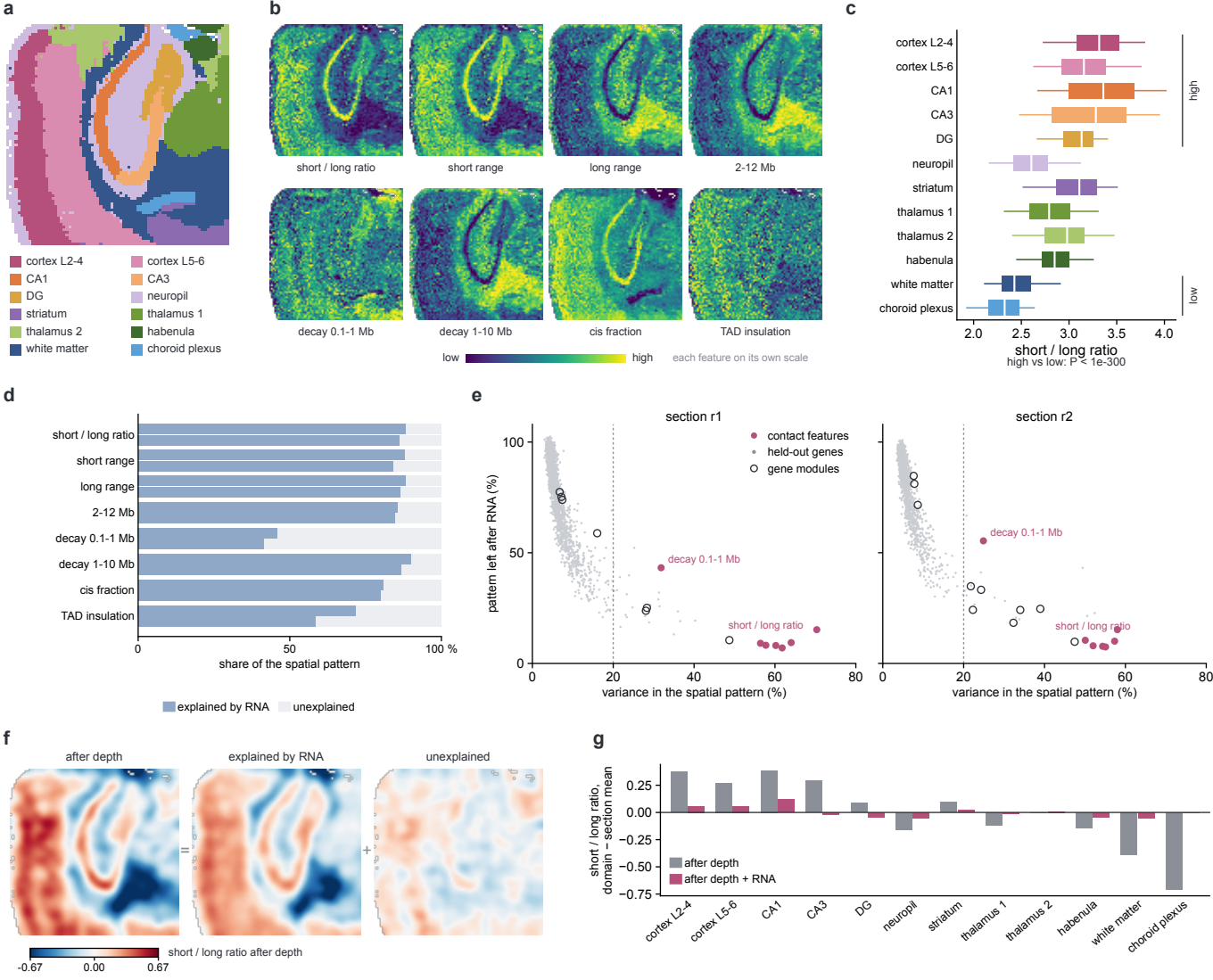

a
cortex L2-4
cortex L5-6
CA1
CA3
DG
neuropil
striatum
thalamus 1
thalamus 2
habenula
white matter
choroid plexus
b
short / long ratio
short range
long range
2-12 Mb
decay 0.1-1 Mb
decay 1-10 Mb
cis fraction
TAD insulation
low
high
each feature on its own scale
c
high
low
2.0
2.5
3.0
3.5
4.0
short / long ratio
high vs low: P < 1e-300
d
0
50
100 %
share of the spatial pattern
explained by RNA
unexplained
e
section r1
section r2
contact features
held-out genes
gene modules
pattern left after RNA (%)
variance in the spatial pattern (%)
decay 0.1-1 Mb
short / long ratio
f
after depth
explained by RNA
unexplained
short / long ratio after depth
-0.67
0.00
0.67
g
short / long ratio, domain − section mean
after depth
after depth + RNA